

Given: Due:	Further Maths (Core Pure) HW 12	MARK SCHEME
Chapter 7: Matrix Transformations (invariance)		
1) The matrix $\begin{pmatrix} -0.28 & 0.96 \\ -0.96 & -0.28 \end{pmatrix}$ corresponds to a rotation R in the x-y plane. (i) State the centre of the rotation R, and find the angle of rotation (stating whether it is clockwise or anticlockwise). [3] The transformation S is the rotation R followed by reflection in the x-axis. (ii) Write down the matrix corresponding to reflection in the x-axis, and show that the matrix corresponding to the transformation S is $\begin{pmatrix} -0.28 & 0.96 \\ 0.96 & 0.28 \end{pmatrix}$. [3] (iii) Show that the invariant points of the transformation S lie on a straight line, and find the equation of this line. [4]		

- (i) The centre of rotation is the origin.

Comparing with the general rotation matrix $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ for an

anticlockwise rotation of θ about the origin gives

$$\cos \theta = -0.28$$

$$\sin \theta = -0.96$$

So θ is in the third quadrant, and therefore $\theta = -106.3^\circ$ (1 d.p.)

The angle of rotation is 106.3° clockwise.

[3]

- (ii) For reflection in the x-axis, the point (1, 0) is mapped to itself, and the point (0, 1) is mapped to the point (0, -1)

So the matrix corresponding to reflection in the x-axis is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

$$S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -0.28 & 0.96 \\ -0.96 & -0.28 \end{pmatrix} = \begin{pmatrix} -0.28 & 0.96 \\ 0.96 & 0.28 \end{pmatrix}$$

[3]

(iii) For invariant points, $\begin{pmatrix} -0.28 & 0.96 \\ 0.96 & 0.28 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\begin{pmatrix} -0.28x + 0.96y \\ 0.96x + 0.28y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} -1.28x + 0.96y \\ 0.96x - 0.72y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Both these equations simplify to $4x = 3y$

so the invariant points all satisfy the equation $4x = 3y$, and so they all lie on a straight line with equation $4x = 3y$.

[4]



2)

The matrix $\begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$ represents a transformation, T .

(i) Find the invariant points for the transformation T . [3]

(ii) T is a transformation called a shear. The *line of shear* is the line of invariant points for the shear. The *factor* of a shear gives the distance a point is moved as a multiple of its perpendicular distance from the line of shear. What is the factor of the shear T ? [4]

$$(i) \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 3x - 2y \\ 2x - y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

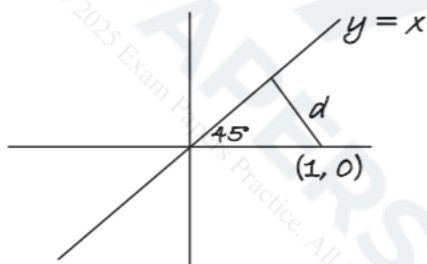
$$\begin{pmatrix} 2x - 2y \\ 2x - 2y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The invariant points are all points on the line $y = x$

[3]

(ii) Consider the point $(1, 0)$.

Under T the point $(1, 0)$ is mapped to the point $(3, 2)$, so the distance it moves is $\sqrt{2^2 + 2^2} = 2\sqrt{2}$.



The perpendicular distance of the point $(1, 0)$ from the line of shear $y = x$ is given by $d = 1 \times \sin 45^\circ = \frac{1}{\sqrt{2}}$.

$$\text{Factor of shear} = \frac{2\sqrt{2}}{1/\sqrt{2}} = 4.$$

[4]

3)

The matrix $\begin{pmatrix} 3 & -2 \\ 4 & -6 \end{pmatrix}$ defines a transformation $\mathbf{M}$ of the (x, y) plane.

(i) Show that the origin is the only invariant point of the transformation. [2]

(ii) Find the two values of m for which $y = mx$ is an invariant line under the transformation. [4]



5. (i) At invariant points, $\begin{pmatrix} 3 & -2 \\ 4 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\begin{pmatrix} 3x - 2y \\ 4x - 6y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$
$$\begin{pmatrix} 2x - 2y \\ 4x - 7y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This can only be true if $x = y = 0$, so the origin is the only invariant point.

[2]

(ii) For a point on the line $y = mx$, image is $\begin{pmatrix} 3 & -2 \\ 4 & -6 \end{pmatrix} \begin{pmatrix} x \\ mx \end{pmatrix} = \begin{pmatrix} 3x - 2mx \\ 4x - 6mx \end{pmatrix}$

For an invariant line, the image point must lie on the line

so $4x - 6mx = m(3x - 2mx)$

$$4 - 6m = 3m - 2m^2$$

$$2m^2 - 9m + 4 = 0$$

$$(2m - 1)(m - 4) = 0$$

$$m = \frac{1}{2} \text{ or } m = 4$$

[4]